

IoE workshop Executive Summary

The Centre for Technology Alternatives for Rural Areas (C-TARA), IIT Bombay, under the Institute of Eminence (IoE), conducted a two-day workshop titled “Experiential Learning in Engineering Through Community Engagement: Agricultural Electricity Distribution Systems (AEDS) ” on 27–28 March 2025. The workshop supported ongoing initiatives under the Unnat Maharashtra Abhiyan (UMA) to strengthen field-based, interdisciplinary engineering education, aligning it with local development challenges. ‘Unnat Maharashtra Abhiyan’ (UMA) is a programme of the Department of Higher and Technical Education, Government of Maharashtra, designed and implemented by the Indian Institute of Technology, Bombay (IITB).

Most engineering courses lack societal, field-based, and real-world experiences, so students are underprepared for their professional roles. Meanwhile, society faces developmental challenges (resource scarcity and sustainability, poorly performing public systems such as the irrigation grid or public transport, farmer incomes, etc.). Now, NEP 2020 has created space for new courses to fulfil the demands of society and industries. In this context, UMA offered a two-day workshop.

The IoE workshop brought together students, faculty from engineering colleges and IIT Bombay, and government officials, fostering rich discussions and collaborative ideas. The focus was to introduce the field-based course and textbook, share pedagogical perspectives, and build a common understanding on integrating experiential learning within engineering curricula as guided by NEP 2020. There was strong support for field-based courses designed to bridge the gap between theory and practice. Faculty members showed enthusiasm for integrating these modules into their curricula. The workshop concluded with a commitment to keep the dialogue going and work together to make engineering education more practical, impactful, and future-ready.

Purpose of the Workshop

The workshop aimed to:

1. Promote field-engaged, interdisciplinary engineering education,
2. Introduce and discuss the field-based course and textbook, including its relevance, design approach, and practical application.
3. Facilitate dialogue between academia and sectoral agencies like MSEDCL, ensuring alignment between educational objectives and real-world needs.
4. Identify pathways for faculty training, inter-institutional collaboration, and knowledge-product development supporting experiential learning.

Overview of the Workshop

The two-day workshop was well attended by a diverse group of 37 participants from academia and Government. Across the two days, the workshop provided participants with a clear understanding of why field-based engineering education is necessary, how the AEDS course has been designed, and how such courses can be implemented within existing academic structures.

Sessions conducted:

- The broader need for experiential learning and the gaps in traditional theory-based engineering education.
- The AEDS course, its field methodology, key modules, and the role of real data from farms, distribution transformers, and agricultural feeders.
- Pedagogical approaches for training engineers to work with community systems, including interdisciplinary perspectives.
- Demonstrations of tools like GIS and OpenDSS for using field data to build engineering models.
- Examples from colleges already implementing field-based courses, showing student interest and practical learning outcomes.
- Discussions on integrating AEDS and similar modules into NEP frameworks through electives, community engagement projects or project-based formats.

Participants also explored strategies for implementation, challenges faced by colleges, faculty capacity needs, and the importance of institutional support. Overall, the workshop offered a coherent roadmap for adopting field-based engineering education supported by UMA and IoE.

Key Themes & Insights

1. Strong demand for field-based courses like AEDS and knowledge products like APAV to bridge the gap between theory and practice.
2. NEP 2020 enables flexible inclusion of such courses through electives, modules, community engagement projects and interdisciplinary offerings.
3. Faculty play a crucial role as facilitators of experiential learning, requiring institutional encouragement and training.
4. Need to develop additional case studies, modules, and examples for motors, solar integration, safety, and system challenges.

Participant Feedback

Participants appreciated the relevance of such a workshop. They noted that:

- Field-based courses improve student understanding and motivation. Such courses strengthen the employability and practical competence of students.
- Faculty require more exposure and systematic training to implement these methods effectively.
- Collaboration with IIT Bombay and other government agencies will be essential for access to data and field sites.

Overall, the response was highly positive, and several institutions committed to adopting AEDS or similar modules.

Follow-Up Actions

To ensure that the outcomes of the IOE Workshop translated into concrete institutional improvements, several follow-up actions were undertaken by participating engineering colleges and UMA. The key thematic actions are as follows:

1. **Integration of Experiential Learning into Curriculum by Regional Institutes:**
Multiple engineering colleges formally incorporated workshop-related content—such as energised irrigation systems and agricultural distribution systems—into their official syllabi for the 2024–25 academic year.
2. **Credit-Bearing Field & Community Engagement Programs:**
Institutions introduced structured field-based components, including 4-month community engineering internships and final-year projects aligned with real distribution and irrigation system challenges, with credit allocations ranging from **3 to 12 credits**.
3. **Strengthening of Practical Skill Development:**
Colleges adopted hands-on technical modules such as **OpenDSS tutorials**, enabling students to practice modelling, simulation, and analysis of rural distribution networks.
4. **Conduct of faculty and student training workshops on field-based learning,**
UMA IIT Bombay conducted various workshops across Maharashtra. UMA is providing support to colleges integrating AEDS from AY 2025–26.

5. Strengthen partnerships with MSEDCL and irrigation departments for field access and data sharing.

Engineering colleges initiated collaborations with MSEDCL and irrigation departments to secure systematic field access, data sharing, and joint problem-solving opportunities.

6. Developing additional thematic modules and knowledge products.

UMA and participating institutes began creating new thematic modules, case studies, and knowledge resources to expand the scope of experiential learning in rural infrastructure systems.

Conclusion

In conclusion, the IoE workshop demonstrated a strong, shared commitment to transforming engineering education through community-engaged, real-world learning. The discussions reaffirmed the importance of experiential learning, the enabling provisions of NEP 2020, and the need for faculty capacity building and institutional support. Participant feedback underscored the relevance of real data, field exposure, and collaboration with sectoral agencies. The subsequent actions undertaken by participating institutions and UMA, including curricular revisions, credit-based field components, technical skill modules, and strengthened partnerships, reflect a clear movement toward community-engaged, real-world engineering education. With continued collaboration and institutional support, these initiatives will help produce engineers who are better prepared, socially sensitive, and capable of addressing Maharashtra's developmental challenges.

Appendix

1. Workshop Brochure:
<https://drive.google.com/file/d/1V3rbIpxx1iJ1Qe-PzOshk6al44Fw3xb-/view?usp=sharing>
2. Minutes of Workshop:
<https://drive.google.com/file/d/1Rb1GVMB4xmMW8jpoG0oWpEAAu52kuvIf/view?usp=sharing>
3. Follow-up with HEIs:
https://drive.google.com/file/d/1mpV9HdmuiOpuJS9Vr24UfWG_WoHWdT4d/view?usp=sharing

